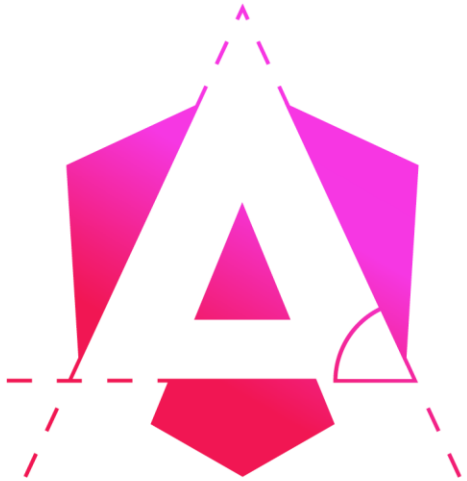

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Angular Performance

Alexander Thalhammer | @LX_T

About me

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About me …

… my office

About me

Hi, it's me → @LX_T https://alex.thalhammer.name/

- **Alex Thalhammer** from Graz, Austria (est. 1983)
  - **Angular Software Tree** GmbH (est. 2019)
- Web Dev for 23 years (I‘ve come a long way baby)
- 2011 - 2017 WordPress Dev (Web, PHP & jQuery)
- Since 2017 **Angular Dev** (Web, TS, Rx - NG 4.0.0)
- Since 2020 **Angular Evangelist, Coach** & **Consultant**
  - Member of **Angular Architects** https://www.angulararchitects.io/

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About you

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Tell us sth about yourself

- Hi, my name is …
- I currently work on …
- A good performance is …
- I have the following problem …
- My goal(s) for this workshop is (are) …

About you

Let us know about your

- Which OS
- Which browser
- Which IDE (IntelliJ/WebStorm vs VS Code)
  - Make sure to update to latest version!!!
- How big is your display?

Pollings

About this workshop

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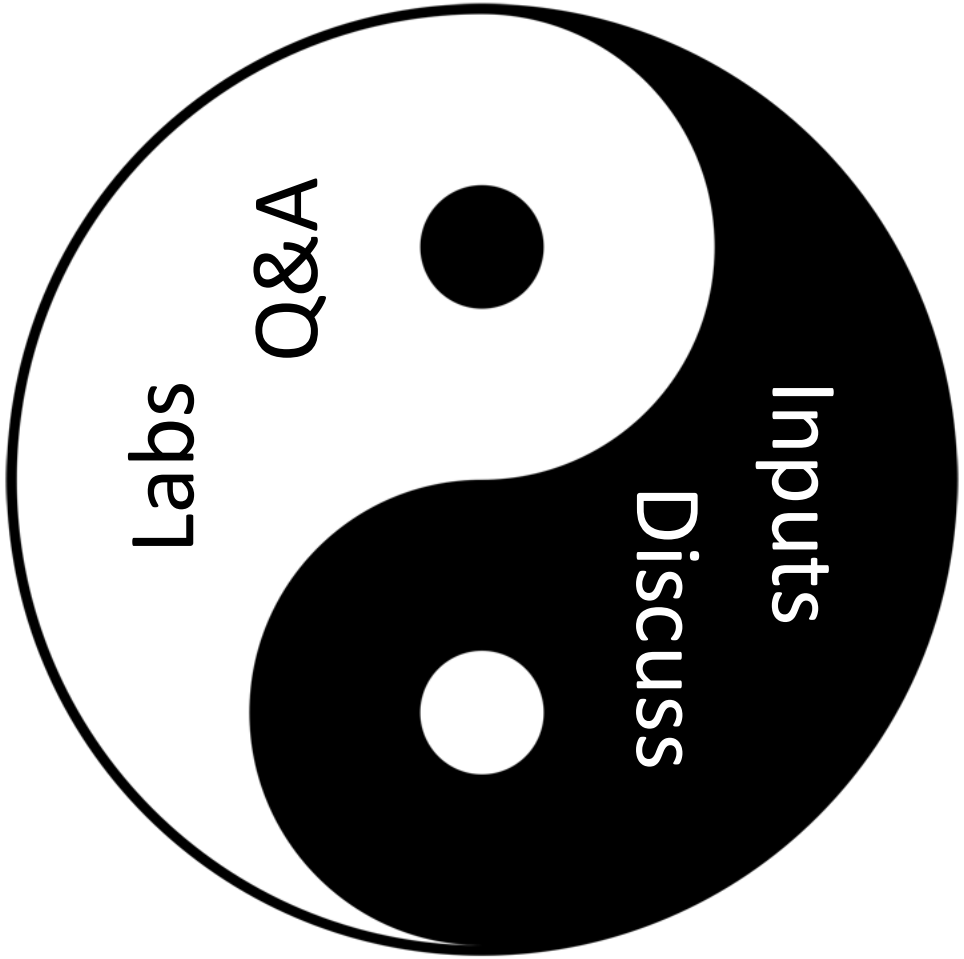
Every participant should

- be able to ***participate*** actively
- learn ***best practices*** in Angular
- improve her/his daily ***work life***
- become a fan of ***Angular***
- have ***fun*** along the way

My goals

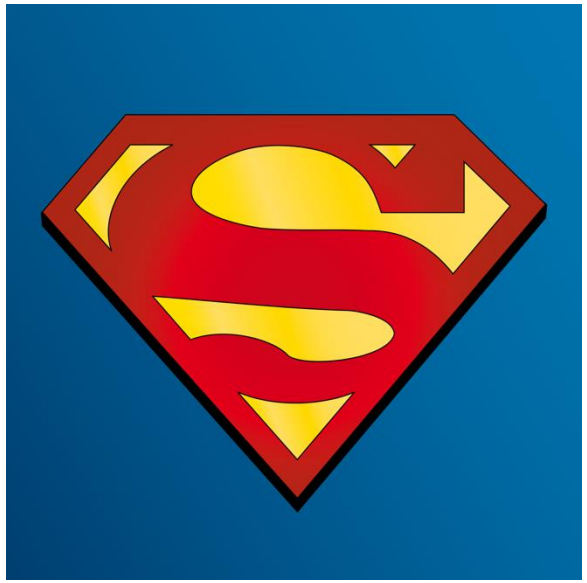


Didactics

We work together I

- Don’t hesitate to interupt me at ***ANY TIME***
  - **Questions**
  - Remarks or
  - Feedback ☺

- Somebody else might have the same **question**
  - Be a hero and ask it



Workshop

We work together II

- Please turn your camera on ***ALL THE TIME***
  - Necessary for me to get your feedback
  - This will also improve the workshop for all ☺

Workshop

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Timetable (approximately)

- 14:00 – 18:00 (incl. 15-20m ☕ break)

- Tomoro: optional Q&A afterwards ☺

Workshop

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What is good performance?

– What did you say?

– **Web performance** refers to the speed in which web pages are downloaded and displayed on the user's web browser. **Web performance optimization (WPO)**, or **website optimization** is the field of knowledge about increasing web performance.
-- https://en.wikipedia.org/wiki/Web_performance

Why Performance Optimization?

– According to *Amazon*, 0.1s less loading time results in around 1% more sales

– According to *Google*, 40% of visitors will abandon a website if it takes longer than 3 seconds to load

– Also according to *Facebook*, when people have to wait too long for a webpage to load, they're more likely to abandon the page all together

Angular Performance Optimization

We distinguish between

- Initial load performance (classical web performance)
- Runtime performance (during usage, e.g. scroll frame rate)

Agenda & Labs

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Agenda

01: UAAO Framework

02: Audit Tools

03: Initial Load - Assets & Build

04: Initial Load - Lazy Loading & Deferrable Views

05: Initial Load - SSR & SSG

06: Runtime - Change Detection

07: Runtime – Best Practices

08: Modern NG (14 – 19)

09: Bonus & Conclusions

Labs

00: Getting Started

03: Initial Load - Assets & Build

06: Runtime - Change Detection

01: Web Performance Audit

04: Initial Load - Lazy Loading & Deferring

07: Runtime – Best Practices

02: Build Analysis

05: Initial Load - SSR & SSG

08: Modern NG (14 – 18)

Setup for Labs

- NodeJS version ^18.19.1 or ^**20.11.1** or ^22
  - https://angular.dev/reference/versions#actively-supported-versions

Actively supported versions

This table covers Angular versions under active support.

Angular	Node.js	TypeScript	RxJS
19.0.x	^18.19.1 || ^20.11.1 || ^22.0.0	>=5.5.0 <5.7.0	^6.5.3 || ^7.4.0
18.1.x || 18.2.x	^18.19.1 || ^20.11.1 || ^22.0.0	>=5.4.0 <5.6.0	^6.5.3 || ^7.4.0
18.0.x	^18.19.1 || ^20.11.1 || ^22.0.0	>=5.4.0 <5.5.0	^6.5.3 || ^7.4.0
17.3.x	^18.13.0 || ^20.9.0	>=5.2.0 <5.5.0	^6.5.3 || ^7.4.0
17.1.x || 17.2.x	^18.13.0 || ^20.9.0	>=5.2.0 <5.4.0	^6.5.3 || ^7.4.0
17.0.x	^18.13.0 || ^20.9.0	>=5.2.0 <5.3.0	^6.5.3 || ^7.4.0

Try "node -v"

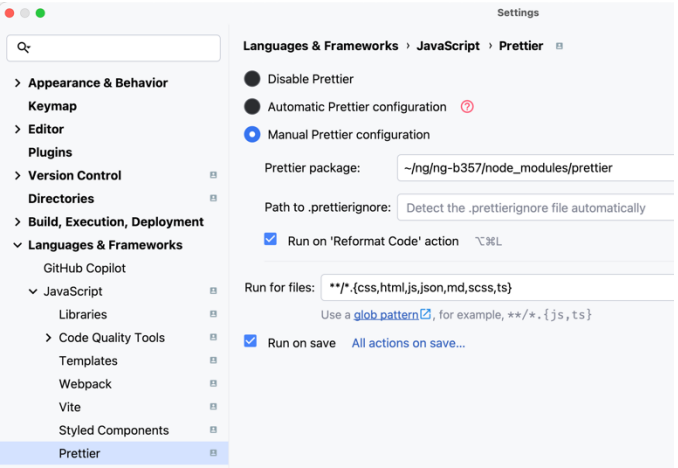
Use NVM ☺

Mac Alternative: Use Homebrew

Setup for Labs

- Git
  - Personally I used to use Sourcetree
    - looking to migrate to Tower | Fork | GitKraken | git extensions
- VS Code (free) | IntelliJ/WebStorm (better)
  - Personally I prefer IntelliJ/WebStorm
- Chrome (or Chromium based alternative)
  - Other browsers are okay for private matters

Recommendations



- Use prettier
  - especially for .ts files
- Use ESLint
  - in your IDE of choice and/or from terminal
- Use Git
  - we recommend to at least commit after every lab
  - easier switching back and forth with my solution

Starter Kit

- Excl. Slides (sent via chat)
- Incl. Labs (exercises)
- Clone https://github.com/L-X-T/ng-b357

Ready for takeoff

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Workshop Updates

- updates are alsways done "on-the-fly"

- NG 18.2
  - Zoneless
  - @let

- NG 19
  - Hybrid Rendering
  - Incremental hydration

Disclaimer

- 00 Intro
- 01 UAAO Framework
- 02 Audit Tools
  - Labs
- 03 Initial Load - Assets & Build
  - Lab
- 04 Initial Load – Lazy Loading
  - Lab

Day 1

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- 05 Initial Load – SSR & SSG
  - Lab
- 06 Runtime - Change Detection
  - Lab
- break
- 07 Runtime - Best Practices
  - Lab
- 08 Modern NG (16-18)
  - Lab
- 09 Bonus & Conclusions

Day 2